Normal Vectors and Quadratic Forms

Linear Algebra · Lecture 10

Last time (Lecture 9): $\mathsf{E}(Az+b) = A\mu + b$, $\operatorname{var}(Az+b) = A\Sigma A'$, $\operatorname{cov}(Az, Bz) = A\Sigma B'$; the 4×4 Σ of Example 5. F1–F5: Chapters 1–6 of the textbook; F6–F11: probability, no proof. $I = I_n$, $e_j = \operatorname{col}_j(I)$.

Facts

F1. $(AB)C = A(BC)$, $A(B+C) = AB + AC$, $(A+B)C = AC + BC$, $c(AB) = (cA)B = A(cB)$, $I_m A = A = AI_n$ ($A \in \mathbb{R}^{m\times n}$); $(A+B)' = A' + B'$, $(cA)' = cA'$, $(AB)' = B'A'$, $(A')' = A$; $x'y = y'x$; A is **symmetric** if $A' = A$.

F2. $(AB)_{ij} = \sum_r a_{ir}b_{rj}$; for $A = [a_1\ \cdots\ a_p]$: $Ae_j = a_j$, $\operatorname{col}_j(AB) = A\operatorname{col}_j(B)$, $AB = a_1b_1' + \cdots + a_pb_p'$ (b_j' = row j of B); $\operatorname{diag}(\alpha_1,\dots,\alpha_n)\operatorname{diag}(\beta_1,\dots,\beta_n) = \operatorname{diag}(\alpha_1\beta_1,\dots,\alpha_n\beta_n)$; block matrices multiply, and are transposed, block by block (when the blocks fit).

F3. $BA = I \Rightarrow B = A^{-1}$; $(A')^{-1} = (A^{-1})'$, so the inverse of an invertible symmetric matrix is symmetric.

F4. $\mathbf{1} = \mathbf{1}_n = (1,\dots,1)' \in \mathbb{R}^n$, $\bar{y} = \frac{1}{n}\mathbf{1}'y$, $\bar{J}_n = \frac{1}{n}\mathbf{1}\mathbf{1}'$, $C_n = I_n - \bar{J}_n$: $C_n' = C_n = C_n^2$, $C_n\mathbf{1} = 0$, $C_n y = y - \bar{y}\mathbf{1}$, $y'C_n y = \sum_i (y_i - \bar{y})^2$.

F5. $\|x\|^2 = x'x$. $Q \in \mathbb{R}^{n\times p}$ has **orthonormal** columns $\iff Q'Q = I_p$. If $p = n$ (Q **orthogonal**): $Q^{-1} = Q'$, $QQ' = I_n$.

F6. (Probability.) A random variable is a number whose value depends on chance. For random variables u, v and numbers a, b, c: $\mathsf{E}(au + bv + c) = a\,\mathsf{E}(u) + b\,\mathsf{E}(v) + c$ (also for finite sums); $\operatorname{var}(u) = \mathsf{E}\big((u - \mathsf{E}(u))^2\big)$; $\operatorname{cov}(u,v) = \mathsf{E}\big((u - \mathsf{E}(u))(v - \mathsf{E}(v))\big)$. $\mathsf{P}(z \in A)$ is the probability that a random vector $z \in \mathbb{R}^n$ takes a value in $A \subseteq \mathbb{R}^n$: $\mathsf{P}(z \in A \cup B) = \mathsf{P}(z \in A) + \mathsf{P}(z \in B)$ for disjoint A, B, and $\mathsf{P}(z \in \mathbb{R}^n) = 1$; y, z have the **same distribution** if $\mathsf{P}(y \in A) = \mathsf{P}(z \in A)$ for all A, and then $\mathsf{E}\,g(y) = \mathsf{E}\,g(z)$. $u_1,\dots,u_m$ are **independent** if $\mathsf{P}(u_1 \in A_1,\dots,u_m \in A_m) = \mathsf{P}(u_1 \in A_1)\cdots\mathsf{P}(u_m \in A_m)$ for all $A_1,\dots,A_m$. If u takes only the values $a_1,\dots,a_m$: $\mathsf{E}\,g(u) = g(a_1)\,\mathsf{P}(u = a_1) + \cdots + g(a_m)\,\mathsf{P}(u = a_m)$. (Sets and functions are those for which the probabilities and expectations are defined.)

F7. Random vectors $u_1 \in \mathbb{R}^{n_1},\dots,u_m \in \mathbb{R}^{n_m}$ whose MGFs (Definition 1) are finite near 0 are independent $\iff M_{(u_1',\dots,u_m')'}(t_1,\dots,t_m) = M_{u_1}(t_1)\cdots M_{u_m}(t_m)$ for all $t_k \in \mathbb{R}^{n_k}$ near 0.

F8. If $u_1,\dots,u_m$ are independent, so are $g_1(u_1),\dots,g_m(u_m)$, for any functions $g_1,\dots,g_m$.

F9. If M_z is finite near 0, its first and second partial derivatives exist there, and at $t = 0$: $\partial M_z/\partial t_i = \mathsf{E}(z_i)$, $\partial^2 M_z/\partial t_j \partial t_i = \mathsf{E}(z_i z_j)$.

F10. If $u_1,\dots,u_m$ are independent: $\mathsf{E}\big(g_1(u_1)\cdots g_m(u_m)\big) = \mathsf{E}\,g_1(u_1)\cdots\mathsf{E}\,g_m(u_m)$ (finite expectations). So independent random vectors u, v have $\operatorname{cov}(u,v) = 0$.

F11. Let $r, \nu, a, b \geqslant 1$. χ^2_r (**chi-square**) is the distribution of $w_1^2 + \cdots + w_r^2$ for independent $w_1,\dots,w_r \sim N(0,1)$; t_ν is that of $Z/\sqrt{V/\nu}$ for independent $Z \sim N(0,1)$, $V \sim \chi^2_\nu$; $F_{a,b}$ is that of $(U/a)/(V/b)$ for independent $U \sim \chi^2_a$, $V \sim \chi^2_b$. For u with one of these distributions, $G(x) = \mathsf{P}(u \leqslant x)$ is continuous, so $\mathsf{P}(u = c) = 0$ for every c; G is strictly increasing on $\mathbb{R}$ (t_ν), or $G(0) = 0$ and G is strictly increasing on $[0,\infty)$ (χ^2_r, $F_{a,b}$). For $\tau \in (0,1)$, the τ **quantile** $t_{\nu,\tau}$, $\chi^2_{r,\tau}$, $F_{a,b,\tau}$ is the unique c with $G(c) = \tau$; so $\mathsf{P}(u > c) = \alpha$ for the $(1-\alpha)$ quantile c.

From Lecture 8 and Lecture 9 (restated; $A \in \mathbb{R}^{n\times n}$).

- Lecture 8, Theorem 9 (A symmetric): (a) $\mathbb{R}^n$ has an orthonormal basis $q_1,\dots,q_n$ of eigenvectors of A. (b) If $q_1,\dots,q_n \in \mathbb{R}^n$ are orthonormal, $Aq_i = \lambda_i q_i$, $Q = [q_1\ \cdots\ q_n]$ and $\Lambda = \operatorname{diag}(\lambda_1,\dots,\lambda_n)$: Q is orthogonal, $A = Q\Lambda Q'$, the λ_i are the eigenvalues of A, and $\operatorname{rank} A = \#\{i : \lambda_i \neq 0\}$. (d) $A = \lambda_1 q_1 q_1' + \cdots + \lambda_n q_n q_n'$.
- Lecture 8, Theorem 11: for symmetric A, $A^2 = A \iff$ every eigenvalue of A is 0 or 1. Example 5: $\operatorname{rank} C_n = n - 1$.

- Lecture 8, Definition 13, Theorem 14(a),(c): a symmetric A is **positive definite** (PD) if $x'Ax > 0$ for all $x \neq 0$, **positive semidefinite** (PSD) if $x'Ax \geqslant 0$ for all x; with λ_i as in Theorem 9(b): PD $\iff$ every $\lambda_i > 0$; PSD $\iff$ every $\lambda_i \geqslant 0$; PD $\Rightarrow$ invertible. Theorem 15(a): B PSD, $K \in \mathbb{R}^{n\times k}$ $\Rightarrow$ $K'BK$ PSD.
- Lecture 9, Definitions 1, 2: for a random vector z, $\mathsf{E}(z) = (\mathsf{E}(z_i))$, $\operatorname{var}(z) = \mathsf{E}\Big((z-\mu)(z-\mu)'\Big)$ ($\mu = \mathsf{E}(z)$) has the entries $\sigma_{ij} = \operatorname{cov}(z_i, z_j)$; $\operatorname{cov}(u,v) = \mathsf{E}\Big((u-\mu_u)(v-\mu_v)'\Big)$ has the entries $\operatorname{cov}(u_i, v_j)$; so $\operatorname{var}\Big((u',v')'\Big)$ has the blocks $\operatorname{var}(u)$, $\operatorname{cov}(u,v)$; $\operatorname{cov}(v,u)$, $\operatorname{var}(v)$.
- Lecture 9, Theorem 3(b),(c): $\operatorname{var}(Az+b) = A\operatorname{var}(z)A'$; $\operatorname{var}(z)$ is PSD. Theorem 4(d): $\mathsf{E}(uu') = \operatorname{var}(u) + \mu_u\mu_u'$.
- Lecture 9, Example 5: for the Σ of Example 5 below, $a = (-1,-2,1,0)'$ and $b = (2,2,-2,1)'$: $\Sigma a = (0,0,1,2)'$, $\Sigma b = (0,0,0,1)'$, $a'\Sigma a = b'\Sigma b = 1$, $\mathbf{1}'\Sigma\mathbf{1} = 51$; $\operatorname{var}(z_1+z_2) = \operatorname{var}(z_1-z_2) = 5$, $\operatorname{cov}(z_1+z_2, z_1-z_2) = 3$ if $\operatorname{var}(z) = \Sigma$.

1. Normal vectors

Definition 1. The **moment generating function** (MGF) of a random vector $z \in \mathbb{R}^n$ is $M_z(t) = \mathsf{E}\exp(t'z)$ ($t \in \mathbb{R}^n$, where finite). Let $\mu \in \mathbb{R}^n$ and $\Sigma \in \mathbb{R}^{n\times n}$ be symmetric and PSD. z has the **multivariate normal distribution** $N_n(\mu,\Sigma)$, $z \sim N_n(\mu,\Sigma)$, if

$$M_z(t) = \exp\Big(t'\mu + \tfrac{1}{2}t'\Sigma t\Big) \qquad \text{for all } t \in \mathbb{R}^n$$

For $n=1$: $N(\mu,\sigma^2)$, with the MGF $\exp(t\mu + \frac{1}{2}\sigma^2t^2)$; $N(0,1)$ is the **standard normal** distribution. u and v are **jointly normal** if $(u',v')'$ is multivariate normal.

Proposition 2. If $z \sim N_n(\mu,\Sigma)$, then $\mathsf{E}(z) = \mu$ and $\operatorname{var}(z) = \Sigma$.

Proof. The exponent $q(t) = t'\mu + \frac{1}{2}t'\Sigma t = \sum_k t_k\mu_k + \frac{1}{2}\sum_k\sum_l \sigma_{kl}t_kt_l$ ($\sigma_{kl} = \sigma_{lk}$) and the chain and product rules give

$$\frac{\partial q}{\partial t_i} = \mu_i + (\Sigma t)_i, \qquad \frac{\partial^2 q}{\partial t_j \partial t_i} = \sigma_{ij}, \qquad \frac{\partial M_z}{\partial t_i} = \Big(\mu_i + (\Sigma t)_i\Big)M_z(t)$$

$$\frac{\partial^2 M_z}{\partial t_j \partial t_i} = \Big(\sigma_{ij} + \Big(\mu_i + (\Sigma t)_i\Big)\Big(\mu_j + (\Sigma t)_j\Big)\Big)M_z(t)$$

At $t=0$, $M_z(0) = 1$: $\mathsf{E}(z_i) = \mu_i$, $\mathsf{E}(z_iz_j) = \sigma_{ij} + \mu_i\mu_j$ (F9). So $\mathsf{E}(zz') = \Sigma + \mu\mu'$ and $\operatorname{var}(z) = \Sigma$ (Lecture 9, Theorem 4(d)). □

Theorem 3. Let $z \sim N_n(\mu,\Sigma)$, and $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$ constant. Then $Az+b \sim N_m(A\mu+b,\ A\Sigma A')$.

Proof.
$A\Sigma A' = (A')'\Sigma(A')$ is symmetric and PSD (F1; Lecture 8, Theorem 15(a), $K = A'$); for $t \in \mathbb{R}^m$:

$$t'(Az+b) = t'b + (A't)'z \implies M_{Az+b}(t) = \exp(t'b)\,M_z(A't) \qquad \text{(F1; F6)}$$

$$M_{Az+b}(t) = \exp\Big(t'b + (A't)'\mu + \tfrac{1}{2}(A't)'\Sigma(A't)\Big) \qquad \text{(Definition 1)}$$

$$= \exp\Big(t'(A\mu+b) + \tfrac{1}{2}t'(A\Sigma A')t\Big) \qquad \text{(F1)}$$

□

Corollary 4. Let $z \sim N_n(\mu,\Sigma)$.

(a) For $1 \leqslant i_1 < \cdots < i_k \leqslant n$: $(z_{i_1},\dots,z_{i_k})' \sim N_k$, with the mean $(\mu_{i_1},\dots,\mu_{i_k})'$ and the submatrix of Σ in the rows and columns $i_1,\dots,i_k$ as covariance matrix. In particular $z_i \sim N(\mu_i,\sigma_{ii})$; and for $z = (z_1',z_2')'$, $z_1 \in \mathbb{R}^p$, $z_2 \in \mathbb{R}^q$, $\Sigma_{11} \in \mathbb{R}^{p\times p}$,

$$\mu = \begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}, \qquad \Sigma = \begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix} \implies z_1 \sim N_p(\mu_1,\Sigma_{11}), \quad z_2 \sim N_q(\mu_2,\Sigma_{22})$$

(b) $a'z \sim N(a'\mu,\ a'\Sigma a)$ for $a \in \mathbb{R}^n$.

Proof. (a) $T = [e_{i_1}\ \cdots\ e_{i_k}]$: $T'z = (z_{i_1}, \ldots, z_{i_k})'$, and $T'\Sigma T$ has the entries $e'_{i_r}\Sigma e_{i_s} = \sigma_{i_r i_s}$ (F2); Theorem 3 with $A = T'$, $b = 0$.

(b) Theorem 3 with $A = a'$, $b = 0$. □

Example 5. Let $z \sim N_4(\mu, \Sigma)$, with $\mu = (1, 0, 2, 2)'$ and Σ of Lecture 9, Example 5:

$$\Sigma = \begin{pmatrix} 4 & 0 & 4 & 0 \\ 0 & 1 & 2 & 2 \\ 4 & 2 & 9 & 6 \\ 0 & 2 & 6 & 9 \end{pmatrix} = KK', \qquad K = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 \\ 0 & 2 & 2 & 1 \end{pmatrix}$$

($x'\Sigma x = \|K'x\|^2 \geqslant 0$: Σ is PSD.) Find the distributions of z_2, $(z_1, z_3)'$, $\mathbf{1}'z$, $e = z_3 - z_1 - 2z_2$, $f = z_4 + 2z_1 + 2z_2 - 2z_3$ and $(z_1 + z_2, z_1 - z_2)'$.

Solution.

Corollary 4; $e = a'z$, $f = b'z$, $a'\Sigma a = b'\Sigma b = 1$, $\mathbf{1}'\Sigma\mathbf{1} = 51$ (Lecture 9, Example 5):

$$z_2 \sim N(0,1), \qquad \begin{pmatrix} z_1 \\ z_3 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}\right) \qquad \text{(Corollary 4(a))}$$

$$\mathbf{1}'z \sim N(5, 51) \qquad \text{((b); } \mathbf{1}'\mu = 1 + 0 + 2 + 2 = 5)$$

$$e \sim N(1, 1) \qquad (a'\mu = -1 - 0 + 2 + 0 = 1)$$

$$f \sim N(0, 1) \qquad (b'\mu = 2 + 0 - 4 + 2 = 0)$$

$A = \left(\begin{smallmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{smallmatrix}\right)$, $Az = (z_1 + z_2, z_1 - z_2)'$; $A\Sigma A' = \operatorname{var}(Az)$ (Lecture 9, Theorem 3(b)), with the entries of Lecture 9, Example 5:

$$A\mu = \begin{pmatrix} 1 + 0 \\ 1 - 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad Az \sim N_2\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}\right) \qquad \text{(Theorem 3)}$$

2. Independence

Theorem 6. (a) Let $z = (u', v')' \sim N_{p+q}(\mu, \Sigma)$, $u \in \mathbb{R}^p$, $v \in \mathbb{R}^q$, $\mu = (\mu'_u, \mu'_v)'$, $\Sigma = \left(\begin{smallmatrix} \Sigma_{uu} & \Sigma_{uv} \\ \Sigma_{vu} & \Sigma_{vv} \end{smallmatrix}\right)$ ($\Sigma_{uu} \in \mathbb{R}^{p\times p}$). Then u, v are independent $\iff$ $\Sigma_{uv} = 0$ ($\iff$ $\operatorname{cov}(u, v) = 0$).

(b) Let $D = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$. Then $z \sim N_n(\mu, D)$ $\iff$ $z_1, \ldots, z_n$ are independent and $z_i \sim N(\mu_i, \sigma_i^2)$. In particular $z \sim N_n(\mu_0\mathbf{1}, \sigma^2 I_n)$ $\iff$ $z_1, \ldots, z_n$ are independent $N(\mu_0, \sigma^2)$ variables (a **random sample** from $N(\mu_0, \sigma^2)$).

(c) Let $z \sim N_n(\mu, \Sigma)$, and $A \in \mathbb{R}^{p\times n}$, $B \in \mathbb{R}^{q\times n}$ constant. Then Az, Bz are independent $\iff$ $A\Sigma B' = 0$.

Proof. (a) $\Rightarrow$: $\Sigma = \operatorname{var}(z)$ (Proposition 2) has the block $\Sigma_{uv} = \operatorname{cov}(u, v)$ (Lecture 9, Definitions 1, 2), which is 0 if u, v are independent (F10).

(a) $\Leftarrow$: $\Sigma_{uv} = 0$, so $\Sigma_{vu} = \Sigma'_{uv} = 0$; for $s \in \mathbb{R}^p$, $t \in \mathbb{R}^q$, by blocks (F2):

$$(s', t')\mu = s'\mu_u + t'\mu_v, \qquad (s', t')\,\Sigma \begin{pmatrix} s \\ t \end{pmatrix} = s'\Sigma_{uu}s + t'\Sigma_{vv}t$$

$$M_z(s, t) = \exp\Big(s'\mu_u + \tfrac{1}{2}s'\Sigma_{uu}s\Big)\exp\Big(t'\mu_v + \tfrac{1}{2}t'\Sigma_{vv}t\Big) = M_u(s)M_v(t) \qquad \text{(Corollary 4(a))}$$

For all s, t: u, v are independent (F7).

(b) $t'Dt = \sum_i \sigma_i^2 t_i^2$ (F2), so D is PSD; $\Rightarrow$ uses $z_i \sim N(\mu_i, \sigma_i^2)$ (Corollary 4(a)), $\Leftarrow$ uses F10; for all $t \in \mathbb{R}^n$:

$$\Rightarrow \quad M_z(t) = \exp\Big(t'\mu + \tfrac{1}{2}t'Dt\Big) = \textstyle\prod_i \exp\Big(t_i\mu_i + \tfrac{1}{2}\sigma_i^2 t_i^2\Big) = \prod_i M_{z_i}(t_i)$$

$$\Leftarrow \quad M_z(t) = \mathsf{E} \textstyle\prod_i e^{t_i z_i} = \prod_i \mathsf{E}\, e^{t_i z_i} = \prod_i \exp\Big(t_i\mu_i + \tfrac{1}{2}\sigma_i^2 t_i^2\Big) = \exp\Big(t'\mu + \tfrac{1}{2}t'Dt\Big)$$

$\Rightarrow$: $z_1, \ldots, z_n$ are independent (F7). $\Leftarrow$: $z \sim N_n(\mu, D)$ (Definition 1).

(c) $G = \left(\begin{smallmatrix} A \\ B \end{smallmatrix}\right) \in \mathbb{R}^{(p+q)\times n}$, $Gz = \Big((Az)', (Bz)'\Big)'$ (F2):

$$Gz \sim N_{p+q}(G\mu, G\Sigma G'), \qquad G\Sigma G' = \begin{pmatrix} A\Sigma A' & A\Sigma B' \\ B\Sigma A' & B\Sigma B' \end{pmatrix} \qquad \text{(Theorem 3; F2)}$$

By (a) with $u = Az$, $v = Bz$: independent $\iff$ $A\Sigma B' = 0$.

□

Remark 7. Joint normality is needed in Theorem 6(a). Let u take the values -1, 0, 1, each with probability $\frac{1}{3}$, and $v = u^2$. By F6, $\mathsf{E}(u) = \mathsf{E}(u^3) = 0$, $\mathsf{E}(u^2) = \frac{2}{3}$, and $\operatorname{cov}(u, v) = \mathsf{E}(u^3) - \mathsf{E}(u)\,\mathsf{E}(u^2) = 0$; but $\mathsf{P}(u = 0,\ v = 1) = 0 \neq \frac{1}{3} \cdot \frac{2}{3} = \mathsf{P}(u = 0)\,\mathsf{P}(v = 1)$.

Example 8. For z of Example 5: are z_1 and $(z_2, z_4)'$ independent? e and $(z_1, z_2)'$? f and $(z_1, z_2, z_3)'$? $z_1 + z_2$ and $z_1 - z_2$?

Solution.

Corollary 4(a) with $i_1, i_2, i_3 = 1, 2, 4$:

$$\begin{pmatrix} z_1 \\ z_2 \\ z_4 \end{pmatrix} \sim N_3 \left(\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 9 \end{pmatrix} \right)$$

$\sigma_{12} = \sigma_{14} = 0$: z_1 and $(z_2, z_4)'$ are independent (Theorem 6(a), $p = 1$, $q = 2$).

Theorem 6(c) with $B = a'$, $B = b'$; $\Sigma a = (0, 0, 1, 2)'$, $\Sigma b = (0, 0, 0, 1)'$ (Lecture 9, Example 5):

$$[I_2 \ \ 0]\,\Sigma a = (0, 0)' \implies e \text{ and } (z_1, z_2)' \text{ independent}, \qquad z_3 = z_1 + 2z_2 + e$$

$$[I_3 \ \ 0]\,\Sigma b = (0, 0, 0)' \implies f \text{ and } (z_1, z_2, z_3)' \text{ independent}, \qquad z_4 = -2z_1 - 2z_2 + 2z_3 + f$$

$\operatorname{cov}(z_1 + z_2, z_1 - z_2) = 3 \neq 0$: $z_1 + z_2$ and $z_1 - z_2$ are not independent (F10).

3. Conditional distributions

Theorem 9. Let $z \sim N_n(\mu, \Sigma)$ be partitioned as in Corollary 4(a), with Σ_{11} PD, and $\Sigma_{22\cdot1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$. Then $\Sigma_{22\cdot1}$ is PSD, and

$$w = z_2 - \mu_2 - \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1) \sim N_q(0, \Sigma_{22\cdot1}), \qquad w \text{ and } z_1 \text{ independent}$$

So $z_2 = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1) + w$. For $a \in \mathbb{R}^p$, $N_q\big(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(a - \mu_1),\ \Sigma_{22\cdot1}\big)$ is the **conditional distribution of z_2 given $z_1 = a$.**

Proof.

Σ_{11}^{-1} exists (Lecture 8, Theorem 14(c)) and is symmetric (F3); $B = \Sigma_{21}\Sigma_{11}^{-1}$, $B' = \Sigma_{11}^{-1}\Sigma_{12}$ ($\Sigma_{21}' = \Sigma_{12}$); $L = \left(\begin{smallmatrix} I_p & 0 \\ -B & I_q \end{smallmatrix}\right)$; blocks (F2):

$$L(z-\mu) = \begin{pmatrix} z_1 - \mu_1 \\ -B(z_1 - \mu_1) + z_2 - \mu_2 \end{pmatrix} = \begin{pmatrix} z_1 - \mu_1 \\ w \end{pmatrix}$$

$$L\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} - B\Sigma_{11} & \Sigma_{22} - B\Sigma_{12} \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ 0 & \Sigma_{22\cdot1} \end{pmatrix}$$

$$L\Sigma L' = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ 0 & \Sigma_{22\cdot1} \end{pmatrix} \begin{pmatrix} I_p & -B' \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} - \Sigma_{11}B' \\ 0 & \Sigma_{22\cdot1} \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22\cdot1} \end{pmatrix}$$

Theorem 3 with $A = L$, $b = -L\mu$:

$$L(z-\mu) \sim N_n(0, L\Sigma L') \implies w \sim N_q(0, \Sigma_{22\cdot1}) \qquad \text{(Corollary 4(a))}$$

$$\Sigma_{22\cdot1} = \text{var}(w) \text{ is PSD} \qquad \text{(Proposition 2; Lecture 9, Theorem 3(c))}$$

The off-diagonal blocks of $L\Sigma L'$ are 0: $z_1 - \mu_1$ and w are independent (Theorem 6(a)), and so are $z_1 = (z_1 - \mu_1) + \mu_1$ and w (F8).

□

The conditional mean is an affine function of a, and $\Sigma_{22\cdot1}$ does not depend on a. With $B = \Sigma_{21}\Sigma_{11}^{-1}$, $B\Sigma_{11}B' = \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{11}\Sigma_{11}^{-1}\Sigma_{12} = \Sigma_{22} - \Sigma_{22\cdot1}$ is PSD (Lecture 8, Theorem 15(a) with Σ_{11}, B' for its B, K); so $\text{var}(c'w) = c'\Sigma_{22\cdot1}c \leqslant c'\Sigma_{22}c = \text{var}(c'z_2)$ for $c \in \mathbb{R}^q$ (Lecture 9, Theorem 3(b)): knowing z_1 can only reduce the variance.

Example 10. For z of Example 5, find the conditional distribution of $(z_3, z_4)'$ given $z_1 = a_1$, $z_2 = a_2$.

Solution.

Theorem 9 with $(z_1, z_2)'$, $(z_3, z_4)'$ as its z_1, z_2, $p = q = 2$; the blocks of Σ (Example 5); $x'\Sigma_{11}x = 4x_1^2 + x_2^2 > 0$ for $x \neq 0$: Σ_{11} is PD:

$$\Sigma_{11} = \text{diag}(4, 1), \qquad \Sigma_{11}^{-1} = \text{diag}(\tfrac{1}{4}, 1), \qquad \Sigma_{21} = \begin{pmatrix} 4 & 2 \\ 0 & 2 \end{pmatrix}, \qquad \Sigma_{21}\Sigma_{11}^{-1} = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$$

$$\Sigma_{22\cdot1} = \begin{pmatrix} 9 & 6 \\ 6 & 9 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 6 & 9 \end{pmatrix} - \begin{pmatrix} 8 & 4 \\ 4 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

The conditional mean (Theorem 9), $\mu_1 = (1, 0)'$, $\mu_2 = (2, 2)'$:

$$\begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} a_1 - 1 \\ a_2 \end{pmatrix} = \begin{pmatrix} a_1 + 2a_2 + 1 \\ 2a_2 + 2 \end{pmatrix}, \qquad N_2\left(\begin{pmatrix} a_1 + 2a_2 + 1 \\ 2a_2 + 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}\right)$$

The first entry agrees with Example 8: $z_3 = z_1 + 2z_2 + e$, $e \sim N(1, 1)$ independent of $(z_1, z_2)'$. Knowing z_1, z_2 reduces the variances of z_3, z_4 from 9, 9 to 1, 5.

4. The chi-square and t distributions

Proposition 11. If $u \sim \chi^2_r$, then $\mathsf{E}(u) = r$.

Proof. u has the distribution of $w_1^2 + \cdots + w_r^2$ (F11), and $\mathsf{E}(w_i^2) = \mathrm{var}(w_i) + \mathsf{E}(w_i)^2 = 1$ (Proposition 2 with $n = 1$; Lecture 9, Theorem 4(d)); so $\mathsf{E}(u) = \mathsf{E}(w_1^2) + \cdots + \mathsf{E}(w_r^2) = r$ (F6). □

Theorem 12. Let $T \sim t_\nu$, $\alpha \in (0,1)$, $c \in \mathbb{R}$. Then $-T \sim t_\nu$, $\mathsf{P}(T < -c) = \mathsf{P}(T > c)$,

$$\mathsf{P}(T < -t_{\nu,1-\alpha}) = \alpha, \qquad \mathsf{P}\Big(-t_{\nu,1-\alpha/2} \leqslant T \leqslant t_{\nu,1-\alpha/2}\Big) = 1 - \alpha, \qquad t_{\nu,1-\alpha/2} > 0$$

Proof.

$T = Z/\sqrt{V/\nu}$, $Z \sim N(0,1)$, $V \sim \chi^2_\nu$ independent (F11); $-Z \sim N(0,1)$ (Theorem 3, $A = -1$), independent of V (F8); $k = t_{\nu,1-\alpha/2}$:

$$\begin{aligned}
-T = (-Z)/\sqrt{V/\nu} &\sim t_\nu && \text{(F11)}\\
\mathsf{P}(T < -c) = \mathsf{P}(-T > c) &= \mathsf{P}(T > c) && \text{(F6: same distribution)}\\
\mathsf{P}(T < -t_{\nu,1-\alpha}) = \mathsf{P}(T > t_{\nu,1-\alpha}) &= \alpha && \text{(F11)}\\
\mathsf{P}(-k \leqslant T \leqslant k) = 1 - \mathsf{P}(T < -k) - \mathsf{P}(T > k) &= 1 - \tfrac{\alpha}{2} - \tfrac{\alpha}{2} && \text{(line 2 with } c = k\text{; F6, F11)}\\
\mathsf{P}(T < 0) = \mathsf{P}(T > 0), \quad \mathsf{P}(T = 0) = 0 &\implies \mathsf{P}(T \leqslant 0) = \tfrac{1}{2} && \text{(line 2 with } c = 0\text{; F6, F11)}\\
\mathsf{P}(T \leqslant 0) = \tfrac{1}{2} < 1 - \tfrac{\alpha}{2} = \mathsf{P}(T \leqslant k) &\implies k > 0 && \text{(F11: increasing)}
\end{aligned}$$

□

5. Standardization and quadratic forms

Proposition 13. Let Σ be PD, $\Sigma = Q\Lambda Q'$ (Lecture 8, Theorem 9), so $\lambda_i > 0$ (Lecture 8, Theorem 14(a)); $\Lambda^{\pm 1/2} = \mathrm{diag}(\lambda_1^{\pm 1/2}, \ldots, \lambda_n^{\pm 1/2})$, $\Sigma^{1/2} = Q\Lambda^{1/2}Q'$, $\Sigma^{-1/2} = Q\Lambda^{-1/2}Q'$. Then $\Sigma^{1/2}$, $\Sigma^{-1/2}$ are symmetric, and

$$\Sigma^{1/2}\Sigma^{1/2} = \Sigma, \qquad \Sigma^{1/2}\Sigma^{-1/2} = I, \qquad \Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I, \qquad \Sigma^{-1/2}\Sigma^{-1/2} = \Sigma^{-1}$$

Proof.

$Q'Q = I = QQ'$ (F5); products of diagonal matrices (F2):

$$\begin{aligned}
(Q\Lambda^{\pm 1/2}Q')' &= Q\Lambda^{\pm 1/2}Q' && \text{(F1; } \Lambda^{\pm 1/2} \text{ diagonal)}\\
\Sigma^{1/2}\Sigma^{1/2} &= Q\Lambda^{1/2}Q'Q\Lambda^{1/2}Q' = Q\Lambda Q' = \Sigma\\
\Sigma^{1/2}\Sigma^{-1/2} &= Q\Lambda^{1/2}\Lambda^{-1/2}Q' = QQ' = I\\
\Sigma^{-1/2}\Sigma\Sigma^{-1/2} &= Q\Lambda^{-1/2}\Lambda\Lambda^{-1/2}Q' = QQ' = I\\
(\Sigma^{-1/2}\Sigma^{-1/2})\Sigma = Q\Lambda^{-1}Q'Q\Lambda Q' = QQ' = I &\implies \Sigma^{-1/2}\Sigma^{-1/2} = \Sigma^{-1} && \text{(F3)}
\end{aligned}$$

□

Theorem 14. Let $z \sim N_n(\mu, \Sigma)$, Σ PD, and $w = \Sigma^{-1/2}(z-\mu)$. Then $w \sim N_n(0, I_n)$: the w_i are independent and $N(0,1)$; $z = \mu + \Sigma^{1/2}w$; and $(z-\mu)'\Sigma^{-1}(z-\mu) = w'w \sim \chi^2_n$.

Proof.

Theorem 3 with $A = \Sigma^{-1/2}$, $b = -\Sigma^{-1/2}\mu$; $\Sigma^{-1/2}$ is symmetric (Proposition 13):

$$\begin{aligned} w \sim N_n\Big(\Sigma^{-1/2}\mu - \Sigma^{-1/2}\mu,\ \Sigma^{-1/2}\Sigma\Sigma^{-1/2}\Big) = N_n(0, I_n) && \text{(Proposition 13)} \\ \implies w_1, \ldots, w_n \text{ independent, } N(0,1) && \text{(Theorem 6(b))} \\ \Sigma^{1/2}w = \Sigma^{1/2}\Sigma^{-1/2}(z-\mu) = z - \mu & \\ w'w = (z-\mu)'\Sigma^{-1/2}\Sigma^{-1/2}(z-\mu) = (z-\mu)'\Sigma^{-1}(z-\mu) \sim \chi^2_n && \text{(F1; F11)} \end{aligned}$$

□

So $\mathsf{P}\Big((z-\mu)'\Sigma^{-1}(z-\mu) \leqslant \chi^2_{n,1-\alpha}\Big) = 1-\alpha$ (F11).

Theorem 15. Let $z \sim N_n(\mu, \sigma^2 I_n)$ ($\sigma > 0$), and $A \in \mathbb{R}^{n\times n}$ symmetric, $A^2 = A$, $\operatorname{rank} A = r \geqslant 1$, $A\mu = 0$. Then $z'Az/\sigma^2 \sim \chi^2_r$.

Proof.

$A = \sum_i \lambda_i q_i q_i'$ (Lecture 8, Theorem 9(a),(d)), $\lambda_i \in \{0,1\}$ (Lecture 8, Theorem 11), $\#\{i : \lambda_i = 1\} = \operatorname{rank} A = r$ (Lecture 8, Theorem 9(b)); number the q_i so that $\lambda_1 = \cdots = \lambda_r = 1$; $Q_1 = [q_1\ \cdots\ q_r]$:

$$\begin{aligned} A = q_1q_1' + \cdots + q_rq_r' = Q_1Q_1', \qquad Q_1'Q_1 = I_r && \text{(F2; F5)} \\ Q_1'\mu = Q_1'Q_1Q_1'\mu = Q_1'A\mu = 0 & \\ u = \tfrac{1}{\sigma}Q_1'z \sim N_r\Big(0, \tfrac{1}{\sigma^2}Q_1'(\sigma^2 I_n)Q_1\Big) = N_r(0, I_r) && \text{(Theorem 3)} \\ z'Az/\sigma^2 = (Q_1'z)'(Q_1'z)/\sigma^2 = u'u = u_1^2 + \cdots + u_r^2 \sim \chi^2_r && \text{(Theorem 6(b); F11)} \end{aligned}$$

□

Theorem 16. Let $z \sim N_n(\mu, \sigma^2 I_n)$, and $A, B \in \mathbb{R}^{n\times n}$ symmetric with $A^2 = A$, $B^2 = B$.

(a) If $AB = 0$, then $z'Az$ and $z'Bz$ are independent.

(b) If $K \in \mathbb{R}^{m\times n}$ and $KA = 0$, then Kz and $z'Az$ are independent.

Proof.

(a) $A'A = A^2 = A$: $z'Az = z'A'Az = (Az)'(Az)$, a function of Az; likewise $z'Bz = (Bz)'(Bz)$; Theorem 6(c), F8:

$$A(\sigma^2 I_n)B' = \sigma^2 AB = 0 \implies Az,\ Bz \text{ independent} \implies z'Az,\ z'Bz \text{ independent}$$

(b) $z'Az = (Az)'(Az)$ as in (a); Theorem 6(c), F8:

$$K(\sigma^2 I_n)A' = \sigma^2 KA = 0 \implies Kz,\ Az \text{ independent} \implies Kz,\ z'Az \text{ independent}$$

□

6. One normal sample

Theorem 17. Let $n \geqslant 2$, $x_1, \ldots, x_n$ independent $N(\mu, \sigma^2)$ variables ($\sigma > 0$), $\bar{x} = \frac{1}{n}\sum_i x_i$, $s^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2$ and $s = \sqrt{s^2}$.

(a) $\bar{x} \sim N(\mu, \sigma^2/n)$.

(b) $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$.

(c) $\bar{x}$ and s^2 are independent.

(d) $\sqrt{n}(\bar{x} - \mu)/s \sim t_{n-1}$.

Proof. $x = (x_1, \ldots, x_n)' \sim N_n(\mu\mathbf{1}, \sigma^2 I_n)$ (Theorem 6(b)), and $(n-1)s^2 = x'C_n x$ (F4).

(a) Corollary 4(b) with $a = \frac{1}{n}\mathbf{1}$: $a'x = \bar{x}$, $a'(\mu\mathbf{1}) = \mu$, $a'(\sigma^2 I_n)a = \sigma^2/n$.

(b) $C_n' = C_n = C_n^2$, $C_n(\mu\mathbf{1}) = 0$ (F4), $\operatorname{rank} C_n = n - 1 \geqslant 1$ (Lecture 8, Example 5):

$$\frac{(n-1)s^2}{\sigma^2} = \frac{x'C_n x}{\sigma^2} \sim \chi^2_{n-1} \qquad \text{(Theorem 15, } A = C_n\text{)}$$

(c) $K = \frac{1}{n}\mathbf{1}'$: $Kx = \bar{x}$, $KC_n = \frac{1}{n}(C_n\mathbf{1})' = 0$ (F1, F4):

$$\bar{x} \text{ and } x'C_n x \text{ independent} \implies \bar{x} \text{ and } s^2 = \tfrac{1}{n-1}x'C_n x \text{ independent} \qquad \text{(Theorem 16(b); F8)}$$

(d) $Z = \sqrt{n}(\bar{x} - \mu)/\sigma \sim N(0,1)$ ((a); Theorem 3), $V = (n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$ ((b)), independent ((c); F8); $\mathsf{P}(V = 0) = 0$ (F11), so $s > 0$ with probability 1:

$$\frac{\sqrt{n}(\bar{x} - \mu)}{s} = \frac{\sqrt{n}(\bar{x} - \mu)/\sigma}{\sqrt{(n-1)s^2/\big((n-1)\sigma^2\big)}} = \frac{Z}{\sqrt{V/(n-1)}} \sim t_{n-1} \qquad \text{(F11)}$$

□

Definition 18. Let observations be modeled as random variables whose distribution depends on parameters, and $\alpha \in (0,1)$. An interval $[L, U]$ whose endpoints $L \leqslant U$ are computed from the observations is a **confidence interval** with level $1-\alpha$ for a parameter $\theta \in \mathbb{R}$ if $\mathsf{P}(L \leqslant \theta \leqslant U) = 1 - \alpha$ for all values of the parameters.

Example 19. In Theorem 17, let $\alpha \in (0,1)$, $T = \sqrt{n}(\bar{x} - \mu)/s$, $V = (n-1)s^2/\sigma^2$. Find confidence intervals with level $1-\alpha$ for μ and for σ^2; compute them for five measurements $2, 4, 5, 6, 8$, modelled as independent $N(\mu, \sigma^2)$ variables; when does 3 lie in the interval for μ?

Solution.

$c = t_{n-1,1-\alpha/2} > 0$, $T \sim t_{n-1}$ (Theorems 12, 17(d)); $s > 0$:

$$\mathsf{P}(-c \leqslant T \leqslant c) = 1 - \alpha \qquad \text{(Theorem 12; all } \mu, \sigma\text{)}$$

$$-c \leqslant \frac{\sqrt{n}(\bar{x} - \mu)}{s} \leqslant c \iff \bar{x} - \frac{cs}{\sqrt{n}} \leqslant \mu \leqslant \bar{x} + \frac{cs}{\sqrt{n}}$$

So $[\bar{x} - cs/\sqrt{n},\ \bar{x} + cs/\sqrt{n}]$ is a confidence interval for μ with level $1-\alpha$ (Definition 18).

$c_1 = \chi^2_{n-1,\alpha/2}$, $c_2 = \chi^2_{n-1,1-\alpha/2}$, $V \sim \chi^2_{n-1}$ (Theorem 17(b)); F11:

$$\mathsf{P}(c_1 \leqslant V \leqslant c_2) = \mathsf{P}(V \leqslant c_2) - \mathsf{P}(V < c_1) = 1 - \tfrac{\alpha}{2} - \tfrac{\alpha}{2} \qquad (\mathsf{P}(V = c_1) = 0)$$

$$\mathsf{P}(V \leqslant 0) = 0 < \tfrac{\alpha}{2} < 1 - \tfrac{\alpha}{2} \implies 0 < c_1 < c_2$$

$$c_1 \leqslant \frac{(n-1)s^2}{\sigma^2} \leqslant c_2 \iff \frac{(n-1)s^2}{c_2} \leqslant \sigma^2 \leqslant \frac{(n-1)s^2}{c_1}$$

So $[(n-1)s^2/c_2,\ (n-1)s^2/c_1]$ is a confidence interval for σ^2 with level $1-\alpha$.

The data, $n = 5$:

$$\bar{x} = \tfrac{25}{5} = 5, \qquad x_i - \bar{x}: \ -3, -1, 0, 1, 3, \qquad s^2 = \tfrac{9+1+0+1+9}{4} = 5$$
$$\tfrac{s}{\sqrt{5}} = 1, \qquad (n-1)s^2 = 20$$
$$\mu: \ \Big[5 - t_{4,1-\alpha/2}, \ 5 + t_{4,1-\alpha/2}\Big], \qquad \sigma^2: \ \Big[20/\chi^2_{4,1-\alpha/2}, \ 20/\chi^2_{4,\alpha/2}\Big]$$
$$\mu = 3: \ \frac{\sqrt{5}(\bar{x} - 3)}{s} = \frac{\sqrt{5} \cdot 2}{\sqrt{5}} = 2, \qquad 3 \in \Big[5 - t_{4,1-\alpha/2}, \ 5 + t_{4,1-\alpha/2}\Big] \iff t_{4,1-\alpha/2} \geqslant 2$$

Practice. The Exercises and Problems of the textbook, Chapters 10 and 11: normal vectors, linear transformations and marginal distributions, independence, conditional distributions, the chi-square, t and F distributions, quadratic forms in normal vectors, and one normal sample.